MILITARY SPECIFICATION

CASE, PROJECTILE BURSTER, XM216

This specification is approved for use within the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements and provides quality assurance provisions for the fabrication and packing of the XM216 burster casing for the Cartridge 60mm WP, XM722.

2. APPLICABLE DOCUMENTS

2.1. Government documents.

2.1.1 Specifications and standards. The following specifications and standards from a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

0-A-51	- Acetone, Technical

MILITARY

MIL-A-48078	- Ammunition, Standard Quality Assurance Provisions, General Specification for

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Armament, Munitions and Chemical Command, Attn: AMSMC-QA, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-810	-	Environmental Test Methods and Engineering Guidelines
MIL-STD-1169	-	Packaging, Packing and Marking for Shipment of Inert Ammunition Components

2.1.2 Other government documents, drawings and publications. The following other government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS

US ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

9390728	-	Casing, Burster, XM216

(Copies of specifications, standards, drawings, publications and other Government documents required by manufacturers in connection with specific aquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitations.

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM	E8	-	Tension Testing of Metallic Materials
ASTM	E 18	-	Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials

(Copies may be purchased directly from the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

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2.3 Order of precedence. In the event of conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General. Materials shall comply with requirements specified on the applicable drawings and specifications.

3.2 Parts. The parts shall comply with all applicable drawings and with all requirements specified in this specification. Parts shall be free of burrs which may interfere with further assembly or function of the cartridge.

3.3 Metal defects. The burster casings shall be free from cracks, splits, bursts, cold shuts, porosity, inclusions, folds, seams and other similiar defects. In addition, the cavity shall be free of burrs, scale, fins, pits or other surface discontinuities.

3.4 Air pressure test. Prior to the application of any protective finish, each burster casing shall withstand an internal air pressure of 200 psi minimum for one minute minimum without evidence of leakage when tested in accordance with 4.5.2.

3.5 Protective coating. Protective coating shall be in accordance with the applicable drawing and the specifications referenced there on. Prior to application of the phenolic baking varnish, all surfaces of the burster case shall be throughly cleaned to remove all traces of grit, oil, wax, grease, rust or other foreign matter.

3.5.1 Coating application. The phenolic baking varnish shall be applied to the interior surfaces of the burster case within 4 hours after final cleaning operations have been completed. The varnish applied to the burster case shall produce an adherent film with no visible pinholes, craters, blisters or uneveness. Varnish shall be applied by a spray method unless otherwise authorized by the procuring activity.

3.5.2 Cured coating. After curing, the varnish coating shall be smooth and continuous and shall not soften or loosen when subjected to the acetone curing test (see 4.4.3.3.1 and 4.5.1.1). The acetone shall comply with the requirments of 0-A-51.

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3.5.3 Salt spray. The cured varnish coating shall withstand exposure to a 5 percent salt spray for 48 hours without evidence of failure when tested in accordance with 4.5.1.2.

3.6 Workmanship. The requirements for workmanship are as shown on the applicable drawing, in the applicable specifications and the following:

3.6.1 Surface finish. The requirements for surface finish are as detailed on the applicable drawing.

3.6.2 Threads. Threads shall be full and undamaged for the entire minimum length as required by the applicable drawing.

3.6.3 Burr. No part shall have a burr which might be injurious to personnel handling the item.

3.6.4 Foreign matter. Parts shall be free of dirt, grease, chips, rust, corrosion, or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.1.1 Responsibility for compliance. All items must meet all requirements of section 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

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4.1.2 Alternate inspection provisions. Alternative inspection procedures, methods or equipment (including tool inspection, statistical process control, substitute sampling plans, etc.) may be used by the contractor when they provide, as a minimum the level of quality assurance required by the provisions specified herein. Prior to applying such alternative inspection procedures, methods or equipment, the contractor shall describe them in a written proposal submitted to the Procuring Contracting Officer for evaluation and approval by the Government. When required, the contractor shall demonstrate that the effectiveness of the proposed alternative(s) is equal to or better than the quality assurance provisions herein. In cases of dispute as to whether certain procedures of the contractor's proposed alternative(s) provide equal assurance, the provisions of this specification shall apply.

4.2 Classification of inspections. The following type of inspection shall be conducted on this item:

a. First Article Inspection

b. Quality Conformance Inspection

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.3.2. The first article sample shall consist of the items as required per Table I. Total quantities are as follows:

Part Description	Drawings	Quantity
Case, Projectile Burster without protective coating,	9390728	10
Case, Projectile Burster, with protective coatng	9390728	10

4.3.2 Inspections to be performed. See MIL-A-48078 and Table I specified herein.

4.3.3 Rejection. See MIL-A-48078.

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TABLE I First article inspection

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-70678 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER SEE BELOW
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Case, Projectile Burster, XM216 (Dwg. 9390728)				
	Examination for Defects	10		3.2	4.4.2.1
	Air Leak Test	10		3.4	4.4.3.1/4.5.2
	Tension Test	2		3.2	4.4.3.2/4.5.3

NOTES:

AMSMC Form 1570, 1 Feb 85 Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078 and the following:

A lot shall contain burster casings from not more than one heat of steel, that have been manufactured by the same equipment in one unchanged process. The lot thus constituted shall be the basis for the selection of samples for the tension tests hereinafter prescribed.

4.4.2 Examination. See MIL-A-48078. Unless otherwise specified in the Classification of Defects and Test Tables, sampling plans and procedures for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.1	Case, Projectile Burster, XM216	3 OF 3	9390728
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Metal defective		100%	3.3	Visual
2	Diameter of press fit surface incorrect		100%	3.2	Gage
3	Surface finish of press fit surface incorrect		100%	3.2	Visual/Gage;6.2
4	Burster fails internal air pressure test		100%	3.4	Gage/4.5.2
5	Presence of reverse taper on press fit surface		100%	3.2	Gage
6	Profile of radius blend with press fit surface		100%	3.2	Gage
7	Air test stamp missing		100%	3.2	Visual
Major					
101	Tension test	4		3.2	Gage/4.5.3
102	Outside diameter of front face		0.40%	3.2	Gage
103	Cavity fails to accept plug gage		100%	3.2	Gage
104	Rear outside diameter of burster casing		0.40%	3.2	Gage
105	Diameter of fuze booster cavity (1.200 + .005)		0.40%	3.2	Gage
106	Largest outside diameter		0.40%	3.2	Gage
107	Outside diameter rear of press fit surface (1.467 -.020)		0.40%	3.2	Gage
108	Length from front face to rear of press fit surface		0.40%	3.2	Gage

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.1	Case, Projectile Burster, XM216 (Cont.)	2 OF 3	9390278
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
109	Minor diameter of thread		0.40%	3.2	Gage
110	Length of fuze housing		0.40%	3.2	Gage
111	Total length		0.40%	3.2	Gage
112	Wall thickness at rear of burster case		0.40%	3.2	Gage
113	Perpendicularity of press fit surface to shoulder		0.40%	3.2	Gage
114	Length of undercut, max.		0.40%	3.2	Gage
115	Pitch diameter of thread		0.40%	3.2	Gage
116	Protective finish missing or improper		0.40%	3.5	Visual
117	Curing test	1/		3.5.2	4.4.3.3.1/4.5.1.1
118	Runout of pitch diameter of thread with press fit surface and front face at two lengths		0.40%	3.2	Gage
119	Hardness test		0.40%	3.2	Gage/4.5.4
120	Length from front face to base of fuze booster cavity		0.40%	3.2	Gage
121	Length from front face to shoulder of press fit surface (.19 -.02)		0.40%	3.2	Gage
122	Inside diameter at front face		0.40%	3.2	Gage
123	Wall thickness at sidewall of burster case (3 places at 120 degree apart)		0.40%	3.2	Gage

NOTES: 1/ One burster case from each 4 hour production thereof, shall be subjected to this test.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.1	Case, Projectile Burster, XM216 (cont)	3 OF 3	9390728
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
124	Runout of rear outside diameter with press fit surface and shoulder of press fit surface	2	0.40%	3.2	Gage
125	Salt spray test			3.5.3	4.4.3.3.2/4.5.1.2
126	Presence of burrs in threads		0.40%	3.2	Visual
127	Corrosion preventive compound missing from press fit area		0.40%	3.2	Visual
128	Distance from front face to O.D. forward of rear O.D. (1.370-.010)		0.40%	3.2	Gage
129	Radii missing or incorrect on O.D.		0.40%	3.2	Gage
Minor					
201	Markings missing or improper		0.65%	3.2	Visual
202	Surface finish improper (except on press fit surface)		0.65%	3.2	Visual/6.2
203	Radii or chamfers missing or incorrect (except on outside diameter)		0.65%	3.2	Visual/Gage;6.2
204	Corrosion preventive compound missing from threads		0.65%	3.2	Visual
205	Thread damage		0.65%	3.6.2	Visual
206	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES:

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4.4.3 Testing.

PRECAUTION: This specification covers sampling and testing with chemical, toxic or explosive materials which are potentially hazardous to personnel. Accordingly, it is emphasized that all applicable safety rules, regulations and procedures must be followed in handling and processing these materials.

4.4.3.1 Air leak test. In order to assess material soundness each burster case shall be subjected to an air leak test in accordance with 4.5.2. Any case which fails to sustain the required pressure without leakage shall be rejected.

4.4.3.2 Tension test. In order to determine compliance with mechanical properties requirements two (hardest and softest) from twenty-five selected casings per lot shall be tested in accordance with 4.4.3.4 Failure of any specimen to achieve the required tensile strength or elongation requirement shall be cause for rejection of the lot represented by that heat of steel.

4.4.3.3 Protective coating.

4.4.3.3.1 Curing test. One burster case from each 4 hour production or fraction thereof, shall be tested in accordance with 4.5.1.1. Failure of the sample to comply with 3.5.2 shall be cause for rejection of the production represented by the sample.

4.4.3.3.2 Salt spray. Two burster cases per lot shall be tested in accordance with 4.5.1.2. Failure of any burster case to comply with 3.5.3 shall be cause for rejection of the lot.

4.4.3.4 Hardness test. Twenty-five randomly selected casings per lot shall be subjected to the hardness test specified in 4.5.4 using a Rockwell Hardness Tester or by any other method approved by the procuring activity to assess uniformity of the product. Hardness values shall be recorded.

4.4.4 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed herein is identified either directly or by reference in the "Paragraph Reference/Inspection Method" column of the Quality Conformance Inspection Tables starting with paragraph 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and 6.4 herein.

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4.5 Methods of inspection.

4.5.1 Protective coating.

4.5.1.1 Curing test. An area of at least 1 square inch on the sample burster case, shall be kept wet with acetone for 2 minutes. The area being tested shall be vigorously rubbed with a rubber gloved thumb and visually inspected for evidence of lifting, blistering, or softening of the varnish coating.

4.5.1.2 Salt spray test. The test shall be conducted using component parts painted at the same time in the same equipment as other deliverable parts. The salt solution, equipment, preparation of test parts, test procedures and drying time shall be in accordance with MIL-STD-810 Method 509.2 (salt fog). After drying, any evidence of paint failure, non-adhesion or corrosion on a surface painted with protective finish shall cause rejection of all parts painted during that operation except rust within 1/32 inch of any edges shall be permitted.

4.5.2 Leak test. (See 4.4.3.1) Parts shall be free of rust and other foreign matter in order not to clog passages and possibly result in false nonleakers. Each burster case shall be mounted in an approved fixture in such a manner that the burster case will be completely submerged in water, and then subjected to the required air pressure for the specified time duration. A burster casing shall be classed defective if evidence of bubbles rising to the surface of the water or appearing on any surface of the casing is observed during the test period.

4.5.3 Tension test. (see 4.4.3.2) After successfully passing the leak test, two randomly chosen tension test specimens as prescribed by ASTM Method E-8 shall be taken from the tube position of each of two casing samples (two specimens per casing).

4.5.4 Hardness test. From each sample selected for testing in accordance with 4.4.3.4, two hardness readings on flat areas (one at front of casing, one at rear of casing) shall be recorded. The test shall be performed using the equipment and procedures prescribed by ASTM Method E-18, except that the minimum dwell time shall be 5 seconds when a Rockwell Hardness Test is used.

5. PACKAGING

5.1 Packaging, packing and marking. Packaging, packing and marking shall be in accordance with the requirements of level C in MIL-STD-1169.

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6. NOTES

6.1 Intended use. The components covered by this specification are intended for assembly to XM722 smoke cartridges for use on the 60mm mortar.

6.2 Visual examination qualification. When compliance with the applicable requirement is in doubt as a result of visual examination, this characteristic may be measured or gaged to determine acceptability. The roughness comparison specimens prescribed by ANSI-B46.1 shall be used as the basis for surface roughness determination.

6.3 Ordering data. See MIL-A-48078.

6.4 Submission of inspection equipment designs for approvals. See MIL-A-48078. Submit designs as required to Commander, U.S. Army, ARDEC, ATTN: AMSMC-QAT-I (D), Bldg 62, Picatinny Arsenal, NJ 07806-5000. This address will be specified on the Contract Data Requirements List, DD form 1423 in the contract. Unless otherwise specified, data item DI-R-1714 will apply.

6.5 Drawings. Drawings listed in Section 2 of this specification under the headings US Army Armament Research Development and Engineering Center (ARDEC) may also include drawings prepared by, and identified by as Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal, Picatinny Arsenal or ARRADCOM drawings. Technical data originally prepared by these activities is now under the cognizance of ARDEC.

6.6 Subject term (key word) listing.

Case, burster
Leak, air
Specimens, tensile
Spray, salt
Varnish, phenolic baking

Custodian:
Army-AR

Preparing activity
Army-AR

(Project 1310-A425)

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NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-70678	CASE, PROJECTILE BURSTER, XM216

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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